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Thermodynamic assessment of the Al–Mg–Sm system

B.R. Jia [a], L.B. Liu [a,*], D.Q. Yi [a], Z.P. Jin [a,b], J.F. Nie [c]

[a] *School of Materials Science and Engineering, Central South University, ChangSha, Hunan 410083, PR China*
[b] *State National Key Laboratory of Powder Metallurgy, Central South University, ChangSha, Hunan 410083, PR China*
[c] *School of Physics and Materials Engineering, Monash University, Clayton Campus, Wellington Road, Clayton, Australia*

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Abstract

The Al–Sm and Mg–Sm systems were thermodynamically optimized using the CALPHAD approach. The phases including Liquid, Bcc, Hcp and Rhom were treated as substitutional solution phases, of which the excess Gibbs energies were formulated with Redlich–Kister polynomial functions. The binary intermetallic compounds were treated as stoichiometric phases. Combining with the thermodynamic parameters of the Al–Mg binary systems cited from literature, the Al–Mg–Sm ternary system was thermodynamic extrapolated. The calculated phase equilibria are in good agreement with the experimental data.
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Keywords: Thermodynamic assessment; Phase diagram; CALPHAD; Al–Mg–Sm

1. Introduction

Magnesium alloys have been attracting many attentions in recent years due to their properties such as low density, high specific strength, good castability. Rare–earth elements are often added in Mg–base alloys in order to further improve the corrosion–resistance, castability, and phase stability under 300 °C. However, the mechanism why rare earths can improve the properties of Mg–base alloys has not been clearly identified. For better understanding the precipitating process and design of alloy compositions, knowledge of phase diagrams and thermodynamic data of these systems are crucially necessary.

In order to understand the effect of Sm in the Mg–Al alloy, the Mg–Sm binary system has been optimized with the available experiment data, and the parameters of Al–Sm were partial modified based on the work of Saccone et al. [1]. Together with the thermodynamic parameters of the Al–Mg binary systems by Liang et al. [2], the phase diagram of the Al–Mg–Sm ternary system has been extrapolated. The extrapolated Al–Mg–Sm phase diagram agrees well with the experimental data by Zheng et al. [3].

* Corresponding author. Tel.: +86 7318877732; fax: +86 7318876692.
E-mail address: Pdc@mail.csu.edu.cn (L.B. Liu).

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2. Information of binary systems in the literature

2.1. The Mg–Sm binary system

The first phase diagram of the Mg–Sm system was supposed by Moffatt [4] on the base of literature. According to the experimental results by Iandelli et al. [5,7], Rokhlin et al. [6] and Kinzhibalo et al. [8], four intermetallic compounds: MgSm[5], Mg_2Sm [6,7], Mg_3Sm [7] and $Mg_{41}Sm_5$ [8] were assumed in the phase diagram. According to the Sm rich part diagram reported by Kato and Copeland [9,10], the γ terminal phase was supposed to undergo a eutectoid decomposition at 24.5 at.% Mg and 612 °C; the maximum solubility of the second component in the γ solution was supposed to be at 32.4 at.% Mg at 780 °C.

Saccone et al. [11] have determined the Mg–Sm system by means of differential thermal analysis, X-ray diffraction, metallography and microprobe analysis. Besides of MgSm, Mg_2Sm, Mg_3Sm and $Mg_{41}Sm_5$, another intermetallic compound Mg_5Sm was found. The maximum value of magnesium solubility in the γ phase was found to be about 34.0 at.% Mg at 785 ± 10 °C and the eutectoid decomposition of the γ phase had been observed at 580 ± 10 °C.

In present work, the result by Saccone et al. [11] have been used in the optimization procedure with data from others only for comparison.

2.2. The Al–Sm binary system

There are considerable discussions in the literature about this system. Based on most recent experimental data [1,12,13] and the enthalpies of formation of the compounds [14], Saccone et al. [1] recently assessed this system thermodynamically. There are nine stable phases: Liquid, Bcc, Rhom, Al_2Sm, Al_3Sm, AlSm, $AlSm_2$, $Al_{11}Sm_3$_H and $Al_{11}Sm_3$_L existed in this binary. In order to fit the experiment phase relation in the Al–Mg–Sm ternary system, thermodynamic parameters of some intermetallic compounds have been modified in present work.

2.3. The Al–Mg binary system

The Al–Mg system was thermodynamically studied by Saunders [15] and Zuo and Chang [16]. Recently, Liang et al. [2] reassessed the central part of the Al–Mg phase diagram by thermodynamic calculation based on experimental investigation. In this work, the thermodynamic parameters by Liang et al. [2] were adopted and the calculated phase diagram is shown in Fig. 1.

2.4. The Mg–Al–Sm ternary system

The available experimental result of the ternary Al–Mg–Sm system is the 673 K isothermal section by Zheng et al. [3]. Subsequently, the phase diagram have been assessed by Okamoto et al. [17]. To keep consistence with Al–Mg and Mg–Sm systems, the Al_8Mg_5 and $Mg_{13}Sm_2$ phases in Ref. [3] were described as Al_3Mg_2 and $Mg_{41}Sm_5$ by Okamoto et al. However, the isothermal section of the ternary system is still inconsistent with the boundary binary systems which are adopted in this work. First, Zheng et al. [3] did not find the ε_$Al_{30}Mg_{23}$ phase at 673 K. But Liang et al. [2] suggested this phase should exist in the Al–Mg–Sm. Furthermore, Gröbner [18] found that the ε_$Al_{30}Mg_{23}$ phase was stable at 673 K when they investigated Al–Mg–Gd system. So in present work, the ε_$Al_{30}Mg_{23}$ phase was treated as a stable phase at 673 K. Second, Zheng et al. [3] found the existence of Al_4Sm and $AlSm_3$ phases in the ternary system. However, the Al_4Sm and $AlSm_3$ phases did not exist in the Al–Sm binary system [1]. In this work, those phases weren't thought to be existed. Third, according to the work of Saccone et al. [11], there are five intermetallic compounds MgSm, Mg_2Sm, Mg_3Sm, Mg_5Sm and $Mg_{41}Sm_5$ in the Mg–Sm system, but only four compounds MgSm, Mg_2Sm, Mg_3Sm and $Mg_{13}Sm_2$ was shown on the boundary of the ternary isothermal section.

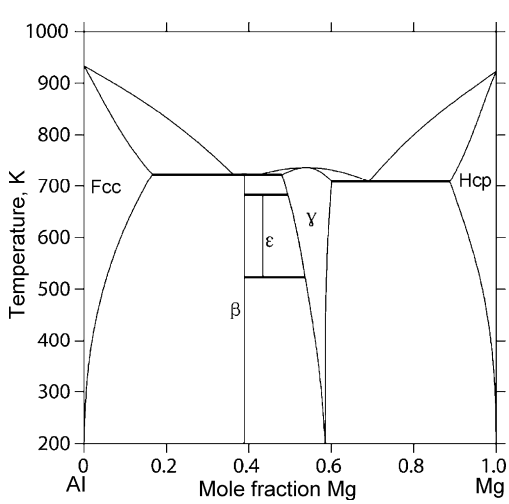

Fig. 1. The calculated Al–Mg phase diagram from [2].

Up to now, there is no experimental or calculated thermochemistry of the Al–Mg–Sm system available in the literature. And no ternary phase has been found in the system. Therefore, the Al–Mg–Sm system has been extrapolated on the basis of the optimized parameters of the three boundary binary systems.

3. Thermodynamic modeling

The Gibbs energies of the solution liquid, Fcc (Al, Mg), Hcp (Mg) and Rhom (Sm) are expressed as:

$$G^{\phi} = \sum_{i=\mathrm{Al,Mg,Sm}} x_i {}^0G_i^{\phi} + RT \sum_{i=\mathrm{Al,Mg,Sm}} x_i \ln(x_i) + {}^{\mathrm{ex}}G^{\phi} \quad (1)$$

Where

$$\begin{aligned} {}^{\mathrm{ex}}G^{\phi} = {} & x_{\mathrm{Al}}x_{\mathrm{Mg}} \sum_{j=0,1,\ldots}^{N} (x_{\mathrm{Al}} - x_{\mathrm{Mg}})^{j(j)} \\ & \times L^{\phi}_{\mathrm{Al,Sm}} + x_{\mathrm{Al}}x_{\mathrm{Sm}} \sum_{j=0,1,\ldots}^{N} (x_{\mathrm{Al}} - x_{\mathrm{Sm}})^{j(j)} \\ & \times L^{\phi}_{\mathrm{Al,Sm}} + x_{\mathrm{Mg}}x_{\mathrm{Sm}} \sum_{j=0,1,\ldots}^{N} (x_{\mathrm{Al}} - x_{\mathrm{Sm}})^{j(j)} \\ & \times L^{\phi}_{\mathrm{Mg,Sm}} + x_{\mathrm{Al}}x_{\mathrm{Mg}}x_{\mathrm{Sm}} L^{\phi}_{\mathrm{Al,Mg,Sm}} \end{aligned} \quad (2)$$

where ϕ denotes the solution phases, x_i (i = Al, Mg and Sm) denotes mole fraction of component i, and ${}^0G_i^{\phi}$ is the molar Gibbs energy of pure element i in the structural state of ϕ and taken from the most recent compilation of Dinsdale [19]. ${}^{(j)}L^{\phi}_{\mathrm{Al,Mg}}$ is taken from Liang et al. [2] and ${}^{(j)}L^{\phi}_{\mathrm{Al,Sm}}$ is taken from Saccone et al.[1]. ${}^{(j)}L^{\phi}_{\mathrm{Mg,Sm}}$ is expressed as:

$${}^{(j)}L^{\phi}_{\mathrm{Mg,Sm}} = A_j + B_j T \quad (3)$$

where the A_j and B_j are constants to be optimized.

$L^{\phi}_{\mathrm{Al,Mg,Sm}}$ is ternary interaction parameters. Due to lack of experiment data, it was not optimized and set to be zero.

In the Al–Mg–Sm ternary system, no ternary intermetallic phase has been found and all the binary intermetallic phases were treated as stoichiometric phases, which were described as M_pN_q. According to Neumann–Kopp rule, the Gibbs energy of the phase M_pN_q was formulated as:

$$G_{\mathrm{MpNq}} = \frac{p}{p+q} {}^0G_{\mathrm{M}}^{\mathrm{HSER}} + \frac{p}{p+q} {}^0G_{\mathrm{N}}^{\mathrm{HSER}} + A + BT \quad (4)$$

Where M and N denote Al, Mg, or Sm, A and B are the adjusted parameters being optimized in the present work.

Table 1
Optimized parameters of the Al–MgvSm ternary system

Phase	Thermodynamic parameters	Ref
Liquid	${}^{0}L^{\mathrm{Liq}}_{\mathrm{Al,Mg}} = -12000.0 + 8.566T$	[16]
	${}^{1}L^{\mathrm{Liq}}_{\mathrm{Al,Mg}} = 1894.0 - 3.0T$	[16]
	${}^{2}L^{\mathrm{Liq}}_{\mathrm{Al,Mg}} = 2000.0$	[16]
	${}^{0}L^{\mathrm{Liq}}_{\mathrm{Al,Sm}} = -168311.85 + 73.4678T$	[1]
	${}^{1}L^{\mathrm{Liq}}_{\mathrm{Al,Sm}} = -42443.15 + 9.8449T$	[1]
	${}^{0}L^{\mathrm{Liq}}_{\mathrm{Mg,Sm}} = -20824.84 + 2.4113T$	This work
	${}^{1}L^{\mathrm{Liq}}_{\mathrm{Mg,Sm}} = -32993.78 + 14.7798T$	This work
Bcc	${}^{0}L^{\mathrm{Bcc}}_{\mathrm{Al,Sm}} = -169345.47 + 79.5101T$	[1]
	${}^{1}L^{\mathrm{Bcc}}_{\mathrm{Al,Sm}} = -41295.02 + 10.3351T$	[1]
	${}^{0}L^{\mathrm{Bcc}}_{\mathrm{Mg,Sm}} = -38189.3587 + 21.4785T$	This work
	${}^{1}L^{\mathrm{Bcc}}_{\mathrm{Mg,Sm}} = -30509.4216 + 19.5937T$	This work
Fcc	${}^{0}L^{\mathrm{Fcc}}_{\mathrm{Al,Mg}} = 4971.0 - 3.5T$	[16]
	${}^{1}L^{\mathrm{Fcc}}_{\mathrm{Al,Mg}} = 900.0 - 0.423T$	[16]
	${}^{2}L^{\mathrm{Fcc}}_{\mathrm{Al,Mg}} = 950.0$	[16]
Hcp	${}^{0}L^{\mathrm{Hcp}}_{\mathrm{Al,Mg}} = 1950.0 - 2.0T$	[16]
	${}^{1}L^{\mathrm{Hcp}}_{\mathrm{Al,Mg}} = 1480.0 - 2.08T$	[16]
	${}^{2}L^{\mathrm{Hcp}}_{\mathrm{Al,Mg}} = 3500.0$	[16]
	${}^{0}L^{\mathrm{Hcp}}_{\mathrm{Mg,Sm}} = 2500 - 3.5T$	This work
Rhom	${}^{0}L^{\mathrm{Rhom}}_{\mathrm{Al,Sm}} = -113981.29 + 60.3948T$	[1]
	${}^{0}L^{\mathrm{Rhom}}_{\mathrm{Mg,Sm}} = -10000 + 5.5T$	This work
βAl_3Mg_2	${}^{0}G^{\beta}_{\mathrm{Al,Mg}} = 0.6114\,{}^{0}G^{\mathrm{Fcc}}_{\mathrm{Al}} + 0.3886\,{}^{0}G^{\mathrm{Hcp}}_{\mathrm{Mg}} - 1075 - 2.95T$	[2]
$\varepsilon Al_{30}Mg_{23}$	${}^{0}G^{\varepsilon}_{\mathrm{Al,Mg}} = 0.566\,{}^{0}G^{\mathrm{Fcc}}_{\mathrm{Al}} + 0.434\,{}^{0}G^{\mathrm{Hcp}}_{\mathrm{Mg}} - 991.8 - 3.2657T$	[2]
$\gamma Al_{12}Mg_{17}$	${}^{0}G^{\gamma}_{\mathrm{Mg:Al:Al}} = 0.8276\,{}^{0}G^{\mathrm{Fcc}}_{\mathrm{Al}} + 0.1724\,{}^{0}G^{\mathrm{Hcp}}_{\mathrm{Mg}} + 3375 - 3.5T$	[2]
	${}^{0}G^{\gamma}_{\mathrm{Mg:Mg:Al}} = 0.4138\,{}^{0}G^{\mathrm{Fcc}}_{\mathrm{Al}} + 0.5862\,{}^{0}G^{\mathrm{Hcp}}_{\mathrm{Mg}} - 1820 - 1.75T$	[2]
	${}^{0}G^{\gamma}_{\mathrm{Mg:Al:Mg}} = 0.4138\,{}^{0}G^{\mathrm{Fcc}}_{\mathrm{Al}} + 0.5862\,{}^{0}G^{\mathrm{Hcp}}_{\mathrm{Mg}} + 9797.4 - 4.761T$	[2]
	${}^{0}G^{\gamma}_{\mathrm{Mg:Mg:Mg}} = {}^{0}G^{\mathrm{Hcp}}_{\mathrm{Mg}} + 4602.4 - 3.011T$	[2]
	${}^{0}L^{\gamma}_{\mathrm{Mg:Al:Al,Mg}} = {}^{0}L^{\gamma}_{\mathrm{Mg:Mg:Al,Mg}} = 3900 - 0.5T$	[2]
$AlSm_2$	${}^{0}G^{\mathrm{AlSm2}}_{\mathrm{Al:Sm}} = 0.3333\,{}^{0}G^{\mathrm{Fcc}}_{\mathrm{Al}} + 0.6667\,{}^{0}G^{\mathrm{Rhom}}_{\mathrm{Sm}} - 38000 + 13.93T$	[1]
	${}^{0}G^{\mathrm{AlSm2}}_{\mathrm{Al:Sm}} = 0.3333\,{}^{0}G^{\mathrm{Fcc}}_{\mathrm{Al}} + 0.6667\,{}^{0}G^{\mathrm{Rhom}}_{\mathrm{Sm}} - 38000 + 13.8T$	This work
AlSm	${}^{0}G^{\mathrm{AlSm}}_{\mathrm{Al:Sm}} = 0.5\,{}^{0}G^{\mathrm{Fcc}}_{\mathrm{Al}} + 0.5\,{}^{0}G^{\mathrm{Rhom}}_{\mathrm{Sm}} - 49000.0 + 16.33T$	[1]
	${}^{0}G^{\mathrm{AlSm}}_{\mathrm{Al:Sm}} = 0.5\,{}^{0}G^{\mathrm{Fcc}}_{\mathrm{Al}} + 0.5\,{}^{0}G^{\mathrm{Rhom}}_{\mathrm{Sm}} - 47500 + 14.95T$	This work
Al_2Sm	${}^{0}G^{\mathrm{Al2Sm}}_{\mathrm{Al:Sm}} = 0.6667\,{}^{0}G^{\mathrm{Fcc}}_{\mathrm{Al}} + 0.3333\,{}^{0}G^{\mathrm{Rhom}}_{\mathrm{Sm}} - 55000 + 15.79T$	[1]
	${}^{0}G^{\mathrm{Al2Sm}}_{\mathrm{Al:Sm}} = 0.6667\,{}^{0}G^{\mathrm{Fcc}}_{\mathrm{Al}} + 0.3333\,{}^{0}G^{\mathrm{Rhom}}_{\mathrm{Sm}} - 55000 + 15.75T$	This work
Al_3Sm	${}^{0}G^{\mathrm{Al3Sm}}_{\mathrm{Al:Sm}} = 0.75\,{}^{0}G^{\mathrm{Fcc}}_{\mathrm{Al}} + 0.25\,{}^{0}G^{\mathrm{Rhom}}_{\mathrm{Sm}} - 48000 + 14.44T$	[1]
	${}^{0}G^{\mathrm{Al3Sm}}_{\mathrm{Al:Sm}} = 0.75\,{}^{0}G^{\mathrm{Fcc}}_{\mathrm{Al}} + 0.25\,{}^{0}G^{\mathrm{Rhom}}_{\mathrm{Sm}} - 49750 + 15.7T$	This work

Table 1 (*Continued*)

Phase	Thermodynamic parameters	Ref
$Al_{11}Sm_3$_H	${}^0G^{Al11Sm3_H}_{Al:Sm} = 0.7857{}^0G^{Fcc}_{Al} + 0.2143{}^0G^{Rhom}_{Sm} - 31153.19 + 4T$	[1]
	${}^0G^{Al11Sm3_H}_{Al:Sm} = 0.7857{}^0G^{Fcc}_{Al} + 0.2143{}^0G^{Rhom}_{Sm} - 31200 + 4T$	This work
$Al_{11}Sm_3$_L	${}^0G^{Al11Sm3_L}_{Al:Sm} = 0.7857{}^0G^{Fcc}_{Al} + 0.2143{}^0G^{Rhom}_{Sm} - 43000 + 12.86T$	[1]
	${}^0G^{Al11Sm3_L}_{Al:Sm} = 0.7857{}^0G^{Fcc}_{Al} + 0.2143{}^0G^{Rhom}_{Sm} - 41785.7 + 11.86T$	This work
MgSm	$G^{MgSm} = 0.5G^{Hcp}_{Mg} + 0.5G^{Rhom}_{Sm} - 10936.2879 + 0.4995T$	This work
Mg_2Sm	$G^{Mg2Sm} = 0.6667G^{Hcp}_{Mg} + 0.3333G^{Rhom}_{Sm} - 10804.9314 + 0.2304T$	This work
Mg_3Sm	$G^{Mg3Sm} = 0.75G^{Hcp}_{Mg} + 0.25G^{Rhom}_{Sm} - 9624.0024 + 0.2313T$	This work
Mg_5Sm	$G^{Mg5Sm} = 0.8333G^{Hcp}_{Mg} + 0.1667G^{Rhom}_{Sm} - 6869.3231 + 0.2989T$	This work
$Mg_{41}Sm_5$	$G^{Mg41Sm5} = 0.8913G^{Hcp}_{Mg} + 0.1087G^{Rhom}_{Sm} - 4496.5746 + 0.013T$	This work

Note: Gibbs energies are expressed in J/mol. Lattice stabilities of elements Mg and Sm are referred to Dinsdale [19].

4. Results and discussion

4.1. The Mg–Sm binary system

On the basis of lattice stabilities cited from Dinsdale [19], the model parameters were optimized using the PARROT module in Thermo–Calc software [20] with experimental phase diagram data. The program works by minimizing an error sum where each of the selected data values is given a certain weight. The weight is chosen by personal judgment and changed by trail and error during the work until most of the selected experimental information is reproduced within the expected uncertainty limits.

All the parameters optimized in the present work are listed in Table 1. The phase diagram calculated with these parameters is depicted in Fig. 2, with reported experimental data. Table 2 indicates that the locations of our calculated invariant phase equilibria are in close agreement with the experiment data of Saccone et al. [11]. An agreement within 10 K between the calculated and experimentally determined temperatures for these invariant reactions has been obtained.

4.2. The Al–Sm binary system

On base of the thermodynamic optimization by Saccone et al. [1], some parameters of the compound phases of the Al–Sm system were modified and listed in Table 1, to fit the phase relations of the experimental isothermal section of the Al–Mg–Sm system by Zheng et al. [3]. The calculated phase diagram was shown in Fig. 3. The assessment enthalpies of formation of the compounds are plotted in Fig. 4. As can be seen, the calculated enthalpies of formation of Al_3Sm, Al_2Sm, AlSm, $AlSm_2$ agree well with the experimental data. The stable temperature range of the $Al_{11}Sm_3$_L is from 538 to 1347 K.

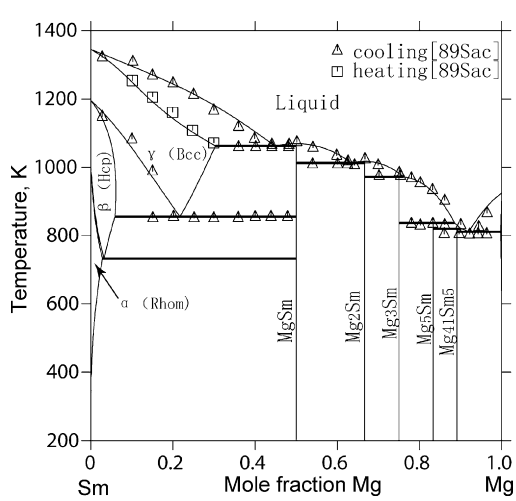

Fig. 2. Comparison of the calculated phase diagram of the Mg–Sm binary system with reported experimental data.

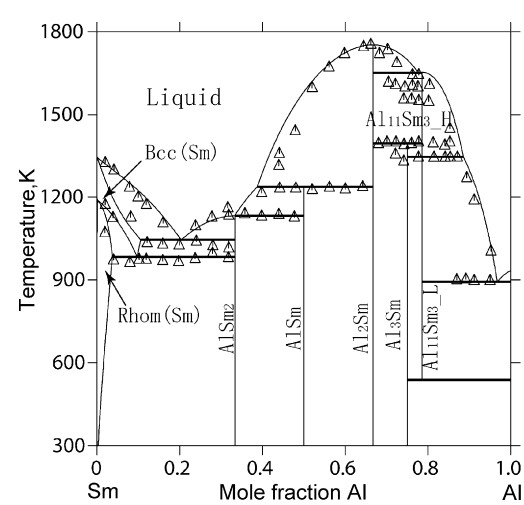

Fig. 3. Calculated phase diagram of the Al–Sm system using parameters from [1] and present work.

Table 2
Invariant reactions in the Mg–Sm system

Reaction	Composition of the phases (at.% Mg)	*T* (K)	Reaction type	Source
L ⇔ Bcc(Sm) + MgSm	45.0, 34.0, 50.0	1058	Eutectic	[11]
	–, 32.4, 50.0	1053	Eutectic	[10]
	44.8, 30.3, 50.0	1063	Eutectic	This work
L ⇔ MgSm	50.0, 50.0	1073	Congruent	[11]
	50.0, 50.0	1069	Congruent	This work
L ⇔ MgSm + Mg_2Sm	64.5, 50.0, 66.7	1008	Eutectic	[11]
	63.4, 50.0, 66.7	1013	Eutectic	This work
L ⇔ Mg_2Sm	66.7, 66.7	1023	Congruent	[11]
	66.7, 66.7	1017	Congruent	This work
L + Mg_2Sm ⇔ Mg_3Sm	77.0, 66.7, 75.0	973	Peritectic	[11]
	76.2, 66.7, 75.0	971	Peritectic	This work
L + Mg_3Sm ⇔ Mg_5Sm	90.0, 75.0, 83.3	838.	Peritectic	[8]
	88.5, 75.0, 83.3	836	Peritectic	This work
L + Mg_5Sm ⇔ $Mg_{41}Sm_5$	90.1, 83.3, 89.1	819	Peritectic	This work
L ⇔ Hcp + $Mg_{41}Sm_5$	92.0, –, 89.1	803	Eutectic	[11]
	92.0, 99.8, 89.1	810	Eutectic	This work
Bcc(Sm) ⇔ Hcp(Sm) + MgSm	20.0, –, 50.0	853	Eutectoid	[11]
	24.5,–, 50.0	885	Eutectoid	[10]
	21.7, 5.86, 50.0	855	Eutectoid	This work
Hcp(Sm) ⇔ Rhom(Sm) + Mgm	3.23, 2.85, 50.0	732	Eutectoid	This work

4.3. The Al–Mg–Sm system

In the Al–Mg–Sm system, there is no ternary compound in the literature. So the phase relations of the Al–Mg–Sm system have been extrapolated based on the optimized parameters of the Al–Sm and Mg–Sm binary systems from the present work and Al–Mg system by Liang et al. [2]. The representative isothermal section with the experimental data at 673 K is illustrated in Fig. 5. The comparison of the calculated results and experimental results is listed in Table 3. Except the intermetallic compound phases Al_4Sm and $AlSm_3$, which have not taken into consideration because of contradict with the Al–Sm binary phase diagram (alloy 1, 2, 12 and 13), the calculated isothermal section have well reproduced the experimental data.

Additionally, the liquids projection with isothermal is further calculated as shown in Fig. 6. The calculated invariant reactions in the Al–Mg–Sm ternary system are summarized in Table 4. Further experimental data are needed to verify the liquids projection.

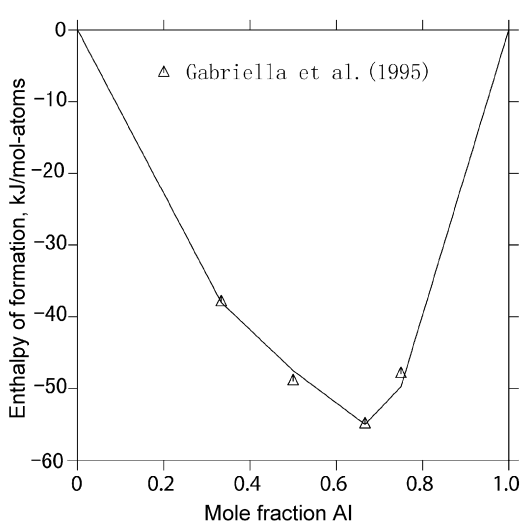

Fig. 4. The calculated enthalpy of formation of the compounds in comparison with various experimental measurements [14].

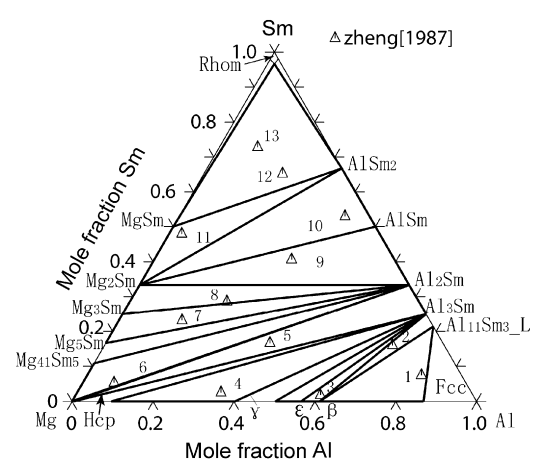

Fig. 5. The extrapolated isothermal section with the experimental data at 673 K.

Table 3
Comparison of calculated and experimental results

Number of three-phase regions	Chemical analysis (at.%)			Results of X-ray analysis [3]	Results of calculated
	Al	Mg	Sm		
1	82.6	9.9	7.5	Al_4Sm,α-Al,Al_3Mg_2	$Al_{11}Sm_3$_L,α-Al,Al_3Mg_2
2	71.2	12.8	16.	Al_4Sm,Al_3Sm,Al_3Mg_2	$Al_{11}Sm_3$_L,Al_3Sm,Al_3Mg_2
3	60.5	37.7	1.8	Al_3Sm,Al_3Mg_2,$Al_{12}Mg_{17}$	Al_3Sm,Al_3Mg_2,$Al_{12}Mg_{17}$
4	35.6	61.9	2.5	Al_3Sm,Mg,$Al_{12}Mg_{17}$	Al_3Sm,Mg,$Al_{12}Mg_{17}$
5	40.6	42.8	16.6	Al_3Sm,Al_2Sm,Mg	Al_3Sm,Al_2Sm,Mg
6	7.8	86.9	5.3	Al_2Sm,Mg,$Mg_{41}Sm_5$	Al_2Sm,Mg,$Mg_{41}Sm_5$
7	15.6	61.2	23.2	Al_2Sm,$Mg_{41}Sm_5$,Mg_3Sm	Al_2Sm,$Mg_{41}Sm_5$,Mg_3Sm
8	24	47.4	28.6	Al_2Sm,Mg_3Sm,Mg_2Sm	Al_2Sm,Mg_3Sm,Mg_2Sm
9	34	25.4	40.6	Al_2Sm,Mg_2Sm,AlSm	Al_2Sm,Mg_2Sm,AlSm
10	41	6.1	52.9	AlSm,$AlSm_2$,Mg_2Sm	AlSm,$AlSm_2$,Mg_2Sm
11	3.1	48.9	48	$AlSm_2$,MgSm,Mg_2Sm	$AlSm_2$,MgSm,Mg_2Sm
12	19.4	15.4	65.2	$AlSm_2$,MgSm,$AlSm_3$	$AlSm_2$,MgSm,Rhom(Sm)
13	9.5	17.7	72.8	$AlSm_3$,MgSm,Sm	$AlSm_2$,MgSm,Rhom(Sm)

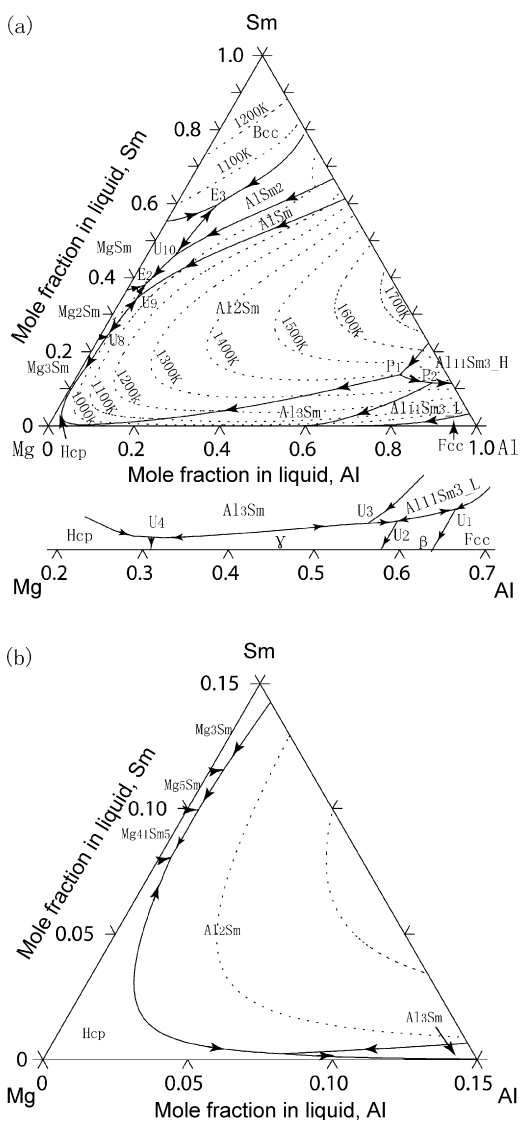

Fig. 6. The calculated liquids projection. (a) Is overview and boundary Mg–Al, (b) is the corner of Mg–rich.

Table 4
Calculated invariant reactions and temperatures

Type	Reaction	T (K)
E_1	$L \Leftrightarrow Hcp(Mg) + Al_2Sm + Mg_{41}Sm_5$	808
E_2	$L \Leftrightarrow Mg_2Sm + AlSm + MgSm$	992
E_3	$L \Leftrightarrow MgSm + AlSm_2 + Bcc(Sm)$	1030
P_1	$L + Al_2Sm + Al_{11}Sm_3_H \Leftrightarrow Al_3Sm$	1396
P_2	$L + Al_3Sm + Al_{11}Sm_3_H \Leftrightarrow Al_{11}Sm_3_L$	1347
U_1	$L + Al_{11}Sm_3_L \Leftrightarrow Fcc(Al) + \beta(Al_3Mg_2)$	724
U_2	$L + Al_{11}Sm_3_L \Leftrightarrow \beta(Al_3Mg_2) + \gamma(Al_{12}Mg_{17})$	723
U_3	$L + Al_3Sm \Leftrightarrow Al_{11}Sm_3_L + \gamma(Al_{12}Mg_{17})$	726
U_4	$L + Al_3Sm \Leftrightarrow Hcp(Mg) + \gamma(Al_{12}Mg_{17})$	709
U_5	$L + Al_2Sm \Leftrightarrow Al_3Sm + Hcp(Mg)$	875
U_6	$L + Mg_5Sm \Leftrightarrow Al_2Sm + Mg_{41}Sm_5$	816
U_7	$L + Mg_3Sm \Leftrightarrow Al_2Sm + Mg_5Sm$	833
U_8	$L + Mg_2Sm \Leftrightarrow Al_2Sm + Mg_3Sm$	961
U_9	$L + Al_2Sm \Leftrightarrow Mg_2Sm + AlSm$	997
U_{10}	$L + AlSm_2 \Leftrightarrow AlSm + MgSm$	1022

5. Conclusion

On the basis of the experimental data, the Mg–Sm system was optimized and the Al–Sm system was modified based on the thermodynamic optimization. The phase relations in the Al–Mg–Sm ternary system at 673 K were reasonably extrapolated. The calculated binary and ternary phase diagrams and thermodynamic data are consistence with experimental data.

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